

IBM HR Analytics Employee Attrition Dataset

Report

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1 Dataset Overview

- **Dataset Description:** The dataset contains comprehensive employee data, including demographics (age, gender), job-specific metrics (department, job role, business travel), compensation details, and employee sentiment (job satisfaction). The target variable is “Attrition,” indicating whether an employee has left the company.
- **Business Domain:** Human Resources (HR) / People Analytics.
- **Objectives of the Analysis:** The primary objective is to uncover the key drivers of employee turnover, identify high-risk employee segments, and provide actionable, data-driven recommendations to HR leadership to improve employee retention.
- **Justification for Selecting the Dataset:** Employee attrition is a major financial and operational burden. This dataset was selected because it provides a realistic, multi-dimensional view of workforce dynamics, allowing for the creation of practical business intelligence solutions that mirror real-world corporate challenges.

2 Data Cleaning

Data Quality Issues Identified:

- Missing values were detected in specific numerical columns (e.g., Daily Rate).
- Duplicate records were found for specific unique employee identifiers (Employee Numbers: 80, 81, 83, 84, 85, and 86).
- Several columns contained zero variance (only a single value across all rows), providing no analytical value.

Cleaning Techniques Performed:

- *Imputation:* Replaced missing numerical values with the mathematical average (mean) to preserve the dataset size.
- *Deduplication:* Removed duplicate rows based on the Employee Number, as this field must act as a unique identifier.
- *Dimensionality Reduction:* Dropped unnecessary columns that offered no analytical insight, specifically: Employee Count (all values were 1), Standard Hours (all values were 80), and Over 18 (all values were ‘Y’).

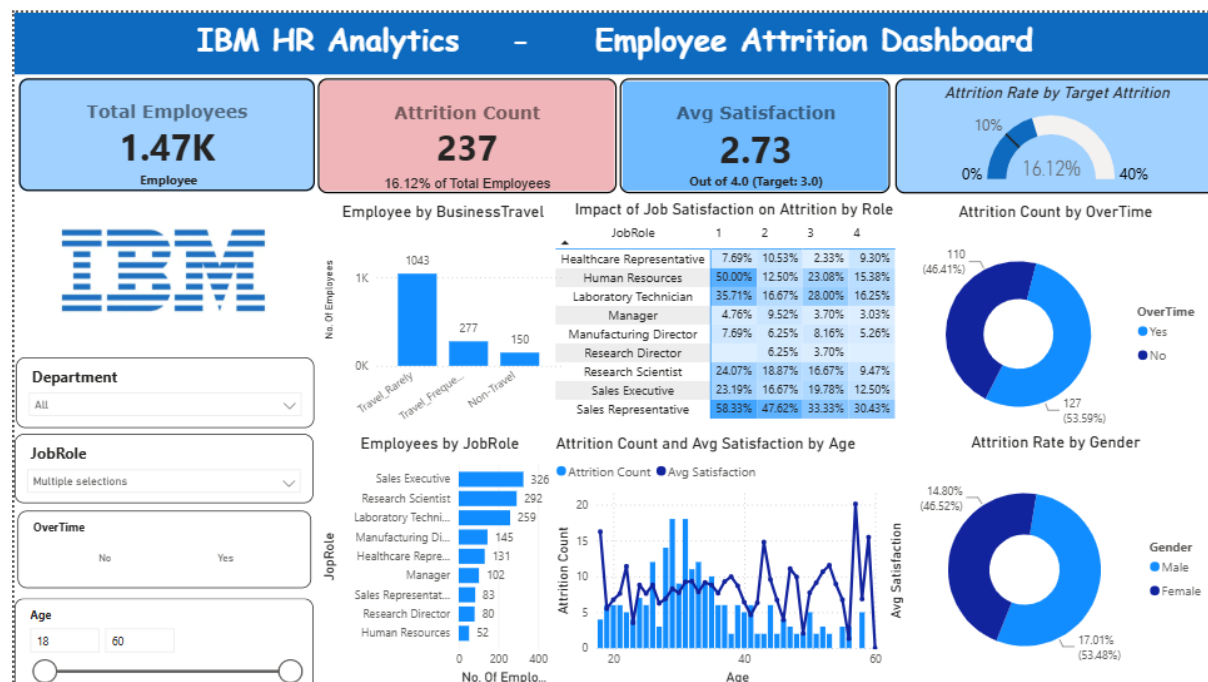
Remaining Limitations:

- Replacing missing values with the average slightly reduces the natural variance of the data. Additionally, survey-based metrics (like Job Satisfaction) are self-reported and may contain inherent subjective bias.

3 Exploratory Data Analysis (EDA)

- **Patterns:** Attrition is heavily skewed toward employees who work regular overtime. There is also a distinct pattern of high turnover in specific entry-level and customer-facing roles.
- **Trends:** A clear trend exists regarding age: younger employees (particularly ages 18–23) show a much higher propensity to leave the company compared to older, more established personnel.
- **Relationships:** A strong negative relationship exists between Job Satisfaction and Attrition. Roles with predominantly low satisfaction scores (Level 1) see severe spikes in turnover.
- **Outliers:** Some outliers were observed in “Years at Company,” with a few veteran employees staying significantly longer than the workforce median, indicating a core group of highly loyal staff amidst higher general turnover.
- **Interesting Observations:** Overtime acts as a massive multiplier for attrition risk, overriding other stabilizing factors like income or department.

4 Dashboard Summary



- **The Purpose of the Dashboard:** The “IBM HR Analytics: Employee Attrition Dashboard” is designed to provide a comprehensive, interactive overview of workforce stability. It enables HR leadership to monitor attrition metrics, evaluate employee satisfaction, and pinpoint high-risk demographics and job roles to facilitate data-driven retention strategies.

- **The Visualizations Used:**

1. *Bar Chart (Employee by BusinessTravel)*: Displays the distribution of the workforce based on travel frequency.
2. *Matrix Table (Impact of Job Satisfaction on Attrition by Role)*: A cross-tabular analysis detailing the exact attrition percentage for each job role across four levels of job satisfaction.
3. *Donut Chart (Attrition Count by OverTime)*: Illustrates the proportion of departed employees who worked overtime versus those who did not.
4. *Bar Chart (Employees by JobRole)*: Ranks job roles by headcount.
5. *Line and Clustered Column Chart (Attrition Count and Avg Satisfaction by Age)*: A dual-axis visualization comparing the absolute number of resignations (bars) against the average job satisfaction score (line) across all employee ages.
6. *Donut Chart (Attrition Rate by Gender)*: Compares the attrition rate between Male and Female employees.

- **How the Dashboard Answers Business Questions:** Through interactive slicers (Department, JobRole, OverTime, and Age), leadership can isolate specific variables. For example, filtering by “OverTime: Yes” instantly recalculates the matrix and age charts, revealing exactly which roles and age groups burn out fastest due to extra hours.

5 KPI Explanation

KPI 1: Total Employees (1.47K)

- **What it measures:** The total active and departed headcount within the analyzed dataset.
- **The target value used:** N/A (This is a baseline volume metric).
- **How the target was selected:** Serves as the denominator for all organizational percentages and rates.
- **Why the chosen target is appropriate:** Essential for contextualizing the scale of the attrition problem.
- **Status:** Provides the foundational scale indicating a mid-to-large enterprise workforce.

KPI 2: Attrition Count (237)

- **What it measures:** The absolute number of employees who have left the organization.
- **The target value used:** The objective is minimization; ideally approaching zero preventable departures.

- **How the target was selected:** Management objective to reduce the raw volume of turnover.
- **Why the chosen target is appropriate:** Tracking the absolute number is vital for calculating the direct financial cost of replacement and training.
- **Status:** A count of 237 indicates a significant loss of organizational knowledge and high upcoming recruitment costs.

KPI 3: Average Satisfaction (2.73)

- **What it measures:** The aggregate job satisfaction score of the workforce, measured on a scale from 1 (Low) to 4 (High).
- **The target value used:** 3.0 (Out of 4.0).
- **How the target was selected:** General HR benchmark denoting a baseline of a “Satisfied” workforce.
- **Why the chosen target is appropriate:** A score of 3.0 acts as a psychological and operational threshold; dropping below this indicates systemic dissatisfaction that precedes mass resignations.
- **Status: Fell below the target.** Sitting at 2.73, the metric indicates widespread employee dissatisfaction, acting as a red flag for future attrition spikes.

KPI 4: Attrition Rate by Target Attrition (16.12%)

- **What it measures:** The percentage of the workforce that has left, visualized against a predefined acceptable limit via a Gauge Chart.
- **The target value used:** 10.00%.
- **How the target was selected:** Standard industry benchmark for healthy organizational turnover.
- **Why the chosen target is appropriate:** A 10% target balances the injection of fresh talent while preventing the excessive loss of experienced personnel.
- **Status: Failed to meet expectations.** The rate exceeded the target by over 6 percentage points (reaching 16.12%), signifying a severe retention crisis that requires immediate intervention.

6 Business Insights and Recommendations

Key Insights:

1. **Overtime is a Primary Catalyst for Turnover:** The donut chart reveals that employees working overtime account for the majority of the attrition count (127 employees, or 53.59%). This suggests that workload and work-life balance are major pain points.

2. **Critical Flight Risk in Specific Roles:** The Matrix visual exposes alarming attrition rates among employees with Level 1 Job Satisfaction in specific departments. Notably, Sales Representatives at Level 1 satisfaction have a massive 58.33% attrition rate, and Human Resources staff at Level 1 show a 50.00% attrition rate.
3. **The Age-Satisfaction Inversion:** The Combo Chart highlights a distinct pattern: attrition counts spike sharply for employees in their late 20s and early 30s. Correspondingly, the average satisfaction line graph dips significantly during these exact age brackets, indicating a "mid-career crisis" phase.
4. **Gender Discrepancy:** Male employees exhibit a slightly higher attrition rate (17.01%) compared to their female counterparts (14.80%), a nuance that can help tailor retention initiatives.

Recommendations:

1. **Restructure Overtime and Workload Policies:** Given that overtime drives the majority of resignations, HR must immediately audit workload distribution. Implementing hard caps on weekly overtime hours or introducing mandatory recuperation days can mitigate burnout and bring the attrition rate closer to the 10.00% target.
2. **Targeted Interventions for Sales and HR:** The 58.33% and 50.00% attrition rates for highly dissatisfied Sales Reps and HR personnel require urgent, localized action. Deploy anonymous pulse surveys and focus groups specifically within these teams to uncover the root causes of their Level 1 satisfaction scores (e.g., compensation, management style, or lack of resources).
3. **Develop Mid-Career Retention Programs:** Address the high turnover among employees in their late 20s and early 30s by introducing clear career progression pathways, leadership development programs, and tailored benefits. Stabilizing satisfaction in this demographic will significantly reduce overall organizational attrition.

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